RUCHITHA MANDA

Austin, Texas | ruchithamanda1199@gmail.com | (848) 278-9878 | Linkedin

SUMMARY

Senior Full-Stack Software Engineer with 6+ years of experience designing, developing, and securing large-scale enterprise applications from modern JavaScript frontends (React.js, Next.js, Angular) to high-performance Java and Spring Boot microservices. Strong experience with REST APIs, service-based architectures, centralized API gateways (Spring Cloud Gateway), and modern authentication (OAuth 2.0/OIDC, JWT, WebAuthn/FIDO2/passkeys, risk-based authentication). Hands-on with testing frameworks (JUnit, Mockito, TestNG, Karma), MongoDB and PostgreSQL, Jenkins and GitHub Actions CI/CD pipelines, code quality enforcement, and full-stack observability. Leverages AI-assisted development (GitHub Copilot, Claude) across the SDLC — including structured prompting and agentic AI workflows — and validates AI-generated output for security and quality.

PROFESSIONAL EXPERIENCE

Senior Software Engineer — Elevance Health, Austin Tx **Feb 2024 – Present**

- Architected 10+ production microservices using Spring Boot, Spring Security, and Spring Data JPA and implemented Outbox patterns for eventual consistency and JWT-based stateless authentication across all service endpoints.
- Implemented a centralized API Gateway using Spring Cloud Gateway with JWT-based authentication, rate limiting, and circuit breaker patterns (Resilience4j), reducing unauthorized access incidents and improving resilience across downstream services.
- Developed secure web applications enforcing authentication and authorization at every layer using Spring Security with OAuth 2.0 and JWT, aligning service-to-service and user-facing flows with enterprise security standards.
- Designed and developed passwordless authentication services using Java, Spring Boot, WebAuthn/FIDO2, and passkeys, enabling users to authenticate through biometrics, security keys, and device-based credentials.
- Built REST APIs for passkey registration, authentication, credential management, and challenge-response validation using Spring Boot and Java; integrated FIDO2/WebAuthn authentication flows with enterprise identity platforms and existing OAuth 2.0/OpenID Connect security frameworks.
- Developed an AI-powered prior authorization assistant using LLM APIs and LangChain to analyze clinical documentation and surface intelligent coverage recommendations, reducing manual administrative review time by ~30% for operations teams.
- Leveraged AI-powered tools (GitHub Copilot, Claude) across the SDLC to accelerate Java/Spring Boot development, automate code generation, testing, documentation and troubleshooting; validated AI-generated code, tests, and documentation against coding and security best practices before merge.
- Migrated enterprise microservices to Java 21 leveraging Virtual Threads to handle high concurrent request throughput with significantly reduced thread management overhead.
- Engineered GraphQL schema federation and RESTful APIs consumed by React.js/Next.js dashboards; implemented SSR and SSG with Next.js, improving page-load performance and boosting organic SEO rankings.
- Built modular, accessible UI components in React.js with Redux state management and functional hooks (useEffect, useContext, useReducer); used Next.js API routes to securely proxy backend service calls.
- Deployed and managed microservices on Red Hat OpenShift by configuring DeploymentConfigs, Pods, Services, Routes, ConfigMaps, Secrets, and Horizontal Pod Autoscalers (HPA), with Spring Boot Actuator health checks, ensuring highly available, scalable and cloud-native application deployments.
- Developed and maintained automated unit & integration test suites using JUnit, Mockito, Spring Boot Test, and Gatling load tests, integrating into CI/CD pipelines to enable automated validation, regression testing, and high-quality production deployments.

Senior Software Engineer — Visa, Austin Tx **Oct 2022 – Dec 2023**

- Developed reusable, enterprise-grade Java/Spring Batch framework with configurable chunk processing, composite item processor, dynamic job, parameter injection, and standardized error handling, reducing new batch job development time by 50% while improving maintainability and consistency across ETL workflows.
- Designed and implemented scalable microservices using Spring Boot and RESTful APIs, driving application modularization, service decoupling, and seamless inter-service communication.
- Implemented risk-based authentication by evaluating device, IP address, location, login history, and behavioral signals to dynamically trigger step-up authentication for high-risk login attempts.
- Integrated authentication services with OAuth 2.0/OIDC, JWT, and Apache Kafka, enabling secure token management, real-time security event processing, and audit logging.
- Integrated OpenAPI/Swagger for automated API documentation, improving developer onboarding, API governance, and cross-team collaboration.

- Built ETL pipelines loading MongoDB data via Apache Spark, transforming and persisting to AWS S3 in Parquet format; optimized Spark jobs through partition pruning and in-memory caching for faster throughput.
- Containerized applications with Docker; orchestrated blue-green deployments on AWS EKS via Kubernetes; automated Jenkins CI/CD pipelines, reducing rollback incidents and overall deployment cycle time.
- Designed and implemented scalable cloud-native solutions on AWS by developing RESTful APIs with AWS API Gateway, building serverless applications using AWS Lambda, integrating SQS and SNS for reliable, event-driven & asynchronous processing and managing AWS infrastructure leveraging EC2, S3, DynamoDB.
- Implemented modular Terraform configurations (modules, workspace, remote state) to provision and manage multi-region AWS infrastructure for multiple microservices.
- Implemented full-stack observability with Prometheus, Grafana, and ELK Stack on Kubernetes clusters; automated infrastructure tasks with Python; integrated Selenium WebDriver for cross-browser UI regression testing.

Associate Engineer — CloudTech, India **Jan 2020 – Jul 2021**

- Built RESTful APIs using Spring Boot, Spring WebFlux, and Spring Data JPA; implemented CQRS event-driven architecture with OAuth 2.0 security, improving application scalability and compliance across all modules.
- Designed Hibernate ORM data models with DAO, DTO, MVC, and Singleton/Factory patterns; wrote complex PL/SQL queries for Oracle data validation; implemented JMS messaging using ActiveMQ and IBM MQ.
- Automated CI/CD pipelines with Jenkins and GitHub Actions; containerized services with Docker; enforced code quality via JUnit, Mockito, and TestNG; developed SOAP/WSDL services using JAX-WS and Apache CXF.
- Developed and maintained Java 8/Spring Boot applications by implementing business requirements, designing RESTful APIs using Spring MVC, and building data access layers with Spring Data JPA/Hibernate, while optimizing SQL queries to enhance application performance and seamless system integration.

EDUCATION

Southern University and A&M College — Master's in Computer and Information Sciences | Baton Rouge, LA

SKILLS

Languages & Frontend: Java, J2EE, JavaScript, TypeScript, Python, SQL, HTML5, CSS3, React.js, Next.js, Angular (v10–16+), NgRx, Redux, Node.js, Express.js, jQuery, AJAX, Karma

Backend & APIs: Spring Boot, Spring MVC, Spring WebFlux, Spring Security, Spring Data JPA, Spring Cloud Gateway, Hibernate ORM, REST, GraphQL, SOAP/WSDL, JAX-WS, Apache CXF, OpenAPI/Swagger

Security & Auth: OAuth 2.0/OIDC, JWT, WebAuthn/FIDO2, Passkeys, Risk-Based Authentication, Spring Security, rate limiting, circuit breakers (Resilience4j), stateless authentication

AI-Assisted Development: GitHub Copilot, Claude, LLM APIs (OpenAI, Azure AI), LangChain, Prompt Engineering, AI-Powered Workflow Automation

Architecture & Patterns: Microservices, Event-Driven Architecture, CQRS, Outbox, Singleton, Factory, Builder, Observer, Repository, MVC, DAO, DTO, Dependency Injection (IoC)

Data & Messaging: PostgreSQL, MongoDB, MySQL, Oracle, MS SQL Server, DynamoDB, Redis Cache, Apache Kafka, JMS (ActiveMQ, IBM MQ), Apache Spark, JDBC

DevOps & CI/CD: Jenkins, GitHub Actions, Docker, Kubernetes, AWS (EC2, S3, Lambda, EKS, API Gateway, SQS, SNS, DynamoDB, CloudWatch, IAM), OpenShift, Terraform, Maven, Nexus, Git, Bitbucket

Testing, Performance & Observability: JUnit, Mockito, TestNG, Jest, Selenium WebDriver, Playwright, Karma, Gatling load testing, Spring Boot Test, Datadog, Splunk, Prometheus, Grafana, ELK Stack, AppDynamics, Spring Boot Actuator, Log4j, Postman, JIRA, Agile/Scrum, TDD